

Money in the Bank: Progressive Problem Shifts in Criminological Science

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ABSTRACT:

Criminology struggles to achieve progressive problem shifts, or cumulative theoretical development common in fields like physics and biology. This stems from a “precision crisis”: a tendency to accept imprecise descriptions, vague theoretical predictions, and analytical practices that obscure data. Drawing on Meehl’s framework, we argue criminological knowledge risks resembling “Monopoly money in the bank” rather than genuine scientific currency. We propose fundamental shifts: prioritizing precise description of concrete phenomena, subjecting theories to risky tests, critically questioning psychometric assumptions, and generating high-resolution data. While academic incentives are a significant barrier, we also explore arguments that these incentives may reflect deeper disciplinary tendencies that favor broader narratives over precise, falsifiable knowledge. The path toward cumulative criminology involves developing approaches that precisely capture complexity while yielding falsifiable knowledge, rather than lowering standards.

KEYWORDS:

precision; scientific progress; Meehl; criminological theory; philosophy of science

CONFLICT OF INTEREST:

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1. Introduction

Despite increased attention to open science reforms (Chin et al., 2023; Sweeten et al., 2024), criminology struggles to achieve consistent progressive problem shifts, or the cumulative theoretical development characteristic of physics or biology. This difficulty stems from a “precision crisis” that manifests not merely in methodological shortcomings but in fundamental assumptions about what forms of knowledge constitute scientific progress. Despite sophisticated methods, criminology often settles for a standardized, reductive knowledge: directional correlations, standardized effect sizes, and significance tests arranged in a theoretical “box score” Meehl (1990b) warned against. The meta-analysis, in this context, culminates this approach, tallying findings onto a simplistic correlation scale. This hegemonic form of knowledge, a historical artifact from an era of limited computational resources, should no longer be social science’s ultimate currency.

This situation reignites deeper critiques of social science research, highlighting a “theory crisis” marked by imprecise constructs, vague predictions, and a failure to distinguish discovery from falsification-oriented testing (Blalock, 1979; Meehl, 1990a; Niemeyer et al., 2025; Oberauer & Lewandowsky, 2019; Robinaugh et al., 2021; Scheel, 2022). Eronen and Bringmann (2021) note psychology’s similar struggles due to a lack of robust phenomena, validity problems, and obstacles to causal discovery. Criminology faces comparable issues (Bruinsma, 2016; Dooley & Goodison, 2020; Ducate et al., 2024).

A foundational figure articulating these challenges was Paul Meehl. In influential papers (1967; 1978; 1990a; 1990b), he argued that social sciences suffer from a pervasive methodological weakness: theories too vague to yield precise, falsifiable predictions, and statistical testing practices (especially “null hypothesis significance testing” or NHST) too weak to challenge these vague theories. Because “everything correlates to some extent” (the “crud factor”), directional predictions almost always find statistical support, making genuine refutation difficult. This contrasts with fields like physics, where theories yield precise quantitative predictions that are genuinely “risky” (Orben & Lakens, 2020).

Meehl characterized this contrast through two powerful metaphors that frame our analysis: progressive sciences accumulate “money in the bank” as genuine, replicable findings that provide a solid foundation for theoretical advancement, while social sciences engage in “box score” science by merely

tallying positive and negative findings without achieving cumulative progress. Recent critics like Mastroianni (2024) have extended this metaphor, likening academic knowledge production to a game of Monopoly where researchers accumulating publications are racing to earn status tokens that create an appearance of scientific value without its substance, or what we might call “Monopoly money in the bank” rather than genuine scientific currency.

Drawing from these critiques, we explore the precision crisis by examining the gap between cumulative science goals and research practices. Following Meehl’s (1986) structure, we organize our discussion around: “Things Criminologists Generally Agree Upon” regarding acknowledged challenges; “Things We Probably Should Understand” concerning the historical contingency of our dominant form of knowledge; and “Things We Don’t Understand” about deeper hurdles requiring transformative approaches to knowledge generation. We aim to identify concrete suggestions for progressive problem shifts, arguing that we must move beyond standardized effect sizes and significance tallies toward richer, more precise descriptions of robust phenomena as the foundation for theory building.

From the outset, we recognize that criminology is diverse, with many questioning whether it constitutes a coherent discipline (Dooley, 2018; Tittle, 2014). Our arguments are rooted in what Tittle (2015) called a “theoretical science” paradigm (see also Turner, 1979), which is characterized by the constant interaction between theory building and testing, valuing detailed measurement, formalized explanation, and anomaly detection. Our suggestions will be most compelling to those who approach criminology from this perspective but may not realize how their quantitative practices, and particularly the forms of knowledge they produce and value, might sometimes contradict the aims of theoretical science.

2. What Should a Good Social Science Theory Look Like?

The struggles of scientific criminology to achieve consistent progressive problem shifts invite a fundamental question: What constitutes a “good” scientific theory, and what is the gap between this ideal and typical practices in our field? Drawing from fields with cumulative success and philosophical accounts of scientific progress, we articulate characteristics of an ideal-typical scientific theory.

A cornerstone of modern philosophy of science, and of quantitative social science in practice, is Popper’s (1959) *falsificationism*. Popper argued that

the hallmark of a scientific theory is its falsifiability, not verifiability. Scientific progress occurs when theories face rigorous attempts at refutation (i.e., “severe tests”). Crucially, Popper emphasized “risky predictions” – specific enough that many potential empirical outcomes would be inconsistent with the theory. If a theory makes only vague predictions consistent with nearly any outcome, it is protected from falsification and cannot be rigorously tested.

Building on Popper, Lakatos (1978) proposed the *methodology of scientific research programs*. Lakatos viewed science not as isolated theories, but as competing research programs, each with a “hard core” of fundamental, irrefutable (by fiat of adherents) theoretical assumptions, protected by a “protective belt” of auxiliary hypotheses and empirical models that are testable. Scientific progress is characterized by *progressive problem shifts*. A research program is “progressive” if its theoretical shifts lead to the discovery of novel facts – predictions of phenomena not previously observed – and if these predictions are empirically confirmed. Progressive programs also offer non-ad-hoc explanations for anomalies that not only explain the anomalies but make new predictions that expand the theory’s explanatory power and potential for falsification. Conversely, a program is “degenerating” or regressive if it only explains known facts (at best), if its auxiliary hypotheses are modified in an ad hoc manner to accommodate anomalies without predicting new phenomena, or if its predictions of novel facts fail. Of course, from a Lakatosian perspective, seemingly degenerating research programs can be subsequently revived if adherents move from ad hocery to progressive shifts. There is always hope.

Drawing on these ideas, we contend that ideal-typical social science theorizing, operating within a theoretical science paradigm, should value and pursue several key characteristics:

2.1. Precision and Formalization.

A good theory is precisely stated, minimizing ambiguity. This precision is crucial for generating risky, falsifiable predictions (Popper, 1959), and where appropriate, is enhanced through formal (e.g., mathematical; statistical) language allowing for unambiguous deductions and predictions (Blalock, 1969; Gibbs, 1987). This contrasts with discursive theories, which often permit multiple interpretations and hinder rigorous falsification.

2.2. Clear, Testable, and Risky Predictions.

Good theories ideally predict not just relationship direction, but approximate magnitude, functional form (e.g., linear; non-linear; threshold), or range under specified conditions (Lakatos, 1978; Popper, 1963). Predictions consistent with almost any data outcome are not risky and preclude rigorous testing or falsification.

2.3. Grounded in and Constrained by Robust Phenomena.

An ideal theory is built upon, and consistently accounts for, a body of precisely described, reproducible empirical regularities – also referred to as “robust phenomena” (Eronen and Bringmann, 2021). These documented patterns in the observable world serve as crucial empirical constraints that limit the plausible forms the theory can take, guiding theoretical development rather than being retrofitted to vague ideas. As an example, consider how Kepler’s precise quantitative description of planetary orbits (Kepler, 1609; discussed in more detail below) constrained Newton’s theory of gravity. Any viable theory had to be able to derive Kepler’s laws.

2.4. Clearly Defined and Empirically Manifest Constructs.

The concepts used are clearly defined. Where they refer to non-observable entities (latent variables), the theory provides a clear rationale for how these constructs should empirically manifest in concrete, measurable indicators. There is a commitment to rigorously validating measures of these constructs and demonstrating that they capture theoretically meaningful variation and exhibit measurement equivalence across relevant contexts and populations (linking to psychometrics and measurement theory; Graham et al., 2025). The relationship between the abstract concept and observable reality is explicit and testable, providing empirical grounding for theoretical claims.

2.5. Specify Processes or Mechanisms.

While avoiding vague “black boxes” or premature reductionism, the theory articulates hypothesized processes, interactions, or mechanisms linking causes to effects (Craver & Darden, 2013; Proctor & Niemeyer, 2021). It specifies the component parts (which could be social, psychological, or situational) and their activities, and clarifies the level of analysis and the scope conditions under which the mechanism operates. The description should be sufficiently detailed to suggest observable implications testable through methods capable of identifying causal effects, contributing to the

protective belt of testable hypotheses in a research program (Lakatos, 1978).

2.6. Explicit Scope Conditions.

The theory clearly delineates the conditions, populations, settings, and time periods under which its predictions are expected to hold (Tittle, 1995). This acknowledges that even robust social science theories may not be universal laws, but clarifies their domain of applicability, making tests more interpretable and predictions more precise under specified conditions. Of course, theories that are more general in scope, and thus more falsifiable, are generally preferred to theories more narrow in scope.

2.7. Progressive in Nature.

The theory builds upon and resolves anomalies from previous research. It predicts novel facts that can guide new, precise research questions and empirical investigations, contributing to a progressive research program that expands the scope of explained phenomena and increases predictive accuracy over time (Lakatos, 1978).

The gap between this ideal and typical criminological theories is substantial. Typical theories are primarily discursive and generate predictions that are often not genuinely risky (Popper, 1959). The crude directional predictions common in the field are easily confirmed by weak statistical tests given the “crud factor” (Meehl, 1990b), hindering falsification and allowing competing theories to persist indefinitely through ad hoc adjustments (Lakatos, 1978; Dooley & Goodison, 2020). Even when potentially progressive problem shifts do occur, severe tests of risky predictions may be quite rare (cf. Brauer & Tittle, 2012; Proctor & Niemeyer, 2020).

Moreover, criminological theories rarely specify magnitudes, functional forms, or interactions with sufficient precision for severe testing (Weak vs. Risky Predictions; Mayo, 2018). They rely heavily on abstract, “lumpy” concepts whose links to concrete measures are often assumed rather than rigorously demonstrated, undermining measurement validity and the precision of theoretical claims (Abstract Constructs with Weak Empirical Links; Blalock, 1979). They often lack systematic grounding in precise quantitative descriptions of fundamental empirical regularities at the level of concrete measures, missing the empirical foundation – the “robust phenomena” (Eronen & Bringmann, 2021) – that guides theory in more cumulative sciences. Finally, scope conditions for a theory’s applicability are often not explicitly stated or rigorously tested.

2.8. Lessons from Progressive Sciences: Concrete Examples of Cumulative Knowledge

To illustrate these principles of good theory in action, consider the following examples from physics and biology that demonstrate progressive problem shifts and cumulative knowledge development:

In astronomy, Johannes Kepler meticulously analyzed Tycho Brahe’s observational data on planetary movements, eventually formulating three laws of planetary motion. His third law specifically stated that the square of a planet’s orbital period is proportional to the cube of its semi-major axis ($T^2 \propto r^3$). This was not merely a correlation or trend, but a precise mathematical relationship that enabled specific, risky predictions: given a planet’s distance from the sun, one could calculate its orbital period with remarkable precision. Later, Newton’s theory of universal gravitation ($F = G(m_1m_2/r^2)$) explained why Kepler’s laws held true, integrating them into a broader theoretical framework while maintaining their predictive precision.

Similarly, in genetics, Gregor Mendel documented precise inheritance patterns in pea plants, identifying that specific traits appeared in exactly predictable ratios (e.g., 3:1 for dominant/recessive traits) across generations. These weren’t merely statistically significant relationships but highly precise, reproducible patterns that allowed for risky predictions: crossing two heterozygous plants would produce offspring with the dominant trait 75% of the time and the recessive trait 25% of the time. These patterns were so precise that deviations demanded explanation rather than being dismissed as noise or heterogeneity. Later, the discovery of DNA’s structure by Watson and Crick provided a mechanism explaining Mendel’s observed patterns, representing a progressive problem shift that expanded explanatory power while maintaining the precise predictions established earlier.

These examples illustrate how progressive sciences build upon precisely described phenomena to develop increasingly powerful theories. Knowledge takes the form of specific, quantitative relationships that enable risky predictions, not merely directional trends or standardized effect sizes. When theories face anomalies, they aren’t protected through ad hoc adjustments but are refined or replaced by theories that explain both the original patterns and the anomalies. This is exemplified by Kepler’s elliptical orbits, which explained observed planetary motions more precisely than earlier models (Ptolemy; Copernicus) that required complex, ad hoc epicycles. This progressive accumulation of increasingly precise and comprehensive explanations represents the kind

of scientific advancement that criminology might aspire to, adapted to its unique subject matter.

In essence, we are arguing that there is a fundamental gap in mainstream criminological knowledge production. The gap lies between a model of science that demands high precision in theoretical claims, rigorous and risky empirical tests that constrain theory and enable falsification, and the cumulative building of knowledge from precisely described empirical foundations, versus a common practice in criminology characterized by relatively vague theories and less demanding empirical tests that are easily satisfied (Meehl, 1990b). This lack of scientific maturity suggests that if theories are too vague to be genuinely falsified by empirical evidence, or if research practices make it too easy to confirm anything, the endeavor may lack the self-correcting mechanism central to scientific progress (Popper, 1959). The following sections delve into how current methodological and theoretical practices contribute to this gap and explore the deeper hurdles and necessary shifts we think are required to propel scientific criminology towards more consistent progressive problem shifts.

3. Things Criminologists Generally Agree Upon: The Landscape of Acknowledged Challenges

Within the theoretical science paradigm, criminologists generally agree on factors complicating research and hindering highly predictive theories. This consensus explains why quantitative progress feels incremental compared to physics or biology. These challenges contribute to criminological models explaining little variance and exhibiting low explanatory and predictive power (Gibbs, 1987; Tittle, 1995; Weisburd & Piquero, 2008), often attributed to the inherent difficulty of the subject matter.

3.1. Complexity and Context-Dependence

Criminology's subject matter – human behavior and social systems – is inherently complex and context-dependent. Unlike the stable particles of classical physics, crime results from dynamic interactions across individual, situational, community, and societal levels, influenced by culture, history, and policy. This complexity precludes simple, universal 'laws' of crime; quantitative theories are thus probabilistic, identifying context-dependent "risk factors" and "correlates" more often than functional causes.

3.2. Data Limitations

Criminologists recognize significant limitations in data availability and quality. While vast amounts

of data exist (e.g., official crime statistics, large-scale surveys), they often suffer from issues such as underreporting, reporting biases, changes in collection methods over time, and a lack of granularity regarding specific details such as timing or motivations of events. Access to certain populations (e.g., active offenders) or sensitive information is often restricted. Relying heavily on administrative data means our measures are often shaped by bureaucratic definitions rather than pure theoretical constructs. While survey data addressed some obstacles (Short & Nye, 1957), funding constraints have yielded mostly moderate- or small-scale sources, with a dearth of large-scale collection suitable for precise study of rare events like serious crime (but see Sampson & Raudenbush, 1999; Hardie & Wikström, 2021). This reliance on fragmented or limited data, coupled with slow data sharing adoption, means the published record often incompletely represents the science.

3.3. Causal Inference Hurdles

Researchers agree on the inherent difficulties in establishing causal relationships in criminology. Many key research questions involve complex social interventions or phenomena that cannot ethically or practically be manipulated through true randomized controlled experiments. While sophisticated quasi-experimental and statistical causal inference methods exist, they rely on strong assumptions that are often difficult to fully meet or definitively test with available data. Disentangling correlation from causation amidst numerous confounding variables remains a persistent challenge (cf. Nagin & Sampson, 2019; Lanfear, Matsueda, & Beach, 2020).

3.4. Perceived Difficulty of Measurement

The very definition and measurement of core concepts in criminology can be challenging and subject to debate. Abstract constructs like "criminal propensity," "social disorganization," or "deterrence perception" are multifaceted, often "lumpy" concepts (as we will discuss), and can be difficult to operationalize consistently and precisely across studies. This perceived difficulty in translating complex theoretical ideas into precise, measurable variables contributes to measurement error, which can attenuate observed relationships and further reduce explained variance in quantitative models.

Cumulatively, these acknowledged challenges – the complexity and context-dependence of the subject matter, data limitations, causal inference hurdles, and the perceived difficulty of measurement – contribute to the widely understood reality of low explanatory and out-of-sample predictive power. However, rather

than viewing this solely as an inevitable consequence of an inherently elusive subject, we contend that focusing on the level of precision we demand in our measurement and analytical practices, and the form of knowledge we produce and value, is crucial for charting a path towards more cumulative quantitative knowledge and achieving progressive problem shifts within a scientific framework.

Despite these challenges, it is important to note that some criminological research has successfully navigated these obstacles and moved toward the ideals outlined in Section 2. Throughout the remainder of this paper, we will cite examples of work that demonstrates elements of the precision and theoretical rigor we advocate. These cases illustrate that higher standards are attainable even within the constraints of studying complex human behavior, providing feasible models for advancing criminological science.

4. Things We Probably Should Understand: The Importance of Accurately Quantifying Complex Phenomena

As discussed in the previous section, scientific criminology faces significant acknowledged challenges. However, low explanatory power isn't solely an inevitable consequence of studying complex phenomena. Instead, much difficulty in achieving cumulative knowledge stems from routinized statistical practices that fail to critically examine historical assumptions about scientific progress in social science. Criminology's dominant quantitative practices reflect a cookbook-like approach where standardized recipes for transforming data into publishable findings are routinely applied with little critical reflection on the form of knowledge generated. Graduate training embeds these routines, prioritizing publication quantity and speed over depth and precision: collect (or download) data, run regression models, report coefficients and *p*-values, interpret direction and significance, and tell a compelling (perhaps post-hoc) causal story about them.

The cookbook approach to quantification is not merely a matter of convenience but reflects historical contingencies in how statistical methods developed and were institutionalized within social science. Methods that simplified complex realities into standardized metrics were revolutionary in their time, enabling analysis in an era of limited computational resources, but have hardened into methodological orthodoxy that constrains the very forms of knowledge we produce and value. This orthodoxy is perpetuated through graduate training, peer review practices, and promotion criteria that reward

adherence to established statistical routines while penalizing methodological innovation that doesn't fit neatly into conventional frameworks of "scientific evidence."

In this section, we examine how these routinized practices emerged historically, how they shape and constrain the forms of knowledge we produce, and how they ultimately hinder the progressive problem shifts necessary for cumulative scientific advancement. Understanding these constraints is essential before we can envision alternatives capable of capturing the complexity of criminological phenomena with the precision needed for robust theory development.

4.1. The Historical Emergence and Limitations of Standardized Effect Sizes

The dominant approach to quantitative criminology, where findings are reduced to correlations, standardized mean differences, and significance tests, emerged from specific historical conditions that no longer constrain us. Early statistical innovations by Pearson, Fisher, and others represented revolutionary advances when calculation was done by hand or rudimentary machines. The correlation coefficient brilliantly compressed complex distributional information into a single number between -1 and +1. Similarly, methods of moments provided computational shortcuts that made statistical inference possible in an era of limited computational resources.

These techniques were genuine innovations that democratized statistical analysis, but their very simplicity may have shackled social scientific inquiry. What began as computational necessity hardened into methodological orthodoxy and eventually shaped our very conception of what scientific knowledge looks like. When our primary statistical currencies are directional correlations and standardized effect sizes, we inadvertently constrain the forms of knowledge we can produce.

For example, the use of the simple correlation coefficient forces complex social phenomena into linear relationships, obscuring potentially crucial non-linear patterns, threshold effects, and context-dependencies. Similarly, our reliance on means and variances (methods of moments) privileges certain distributional features while potentially ignoring others more relevant to theory development. These concerns echo similar critiques in other fields (Gigerenzer, 2018; McShane et al., 2019). Within criminology, some researchers have recognized these limitations and developed approaches that move beyond simplified metrics. For example, Britt and

Weisburd's (2010) work on quantile regression in criminology demonstrated how examining effects across different portions of the outcome distribution reveals patterns that would be obscured by standard regression approaches focused on conditional means.

4.2. Meta-Analysis and the Limits of Current Statistical Practice

Meta-analysis represents both a significant methodological advancement and a reflection of the limitations in our standard statistical practices. While meta-analyses admirably address publication bias and increase statistical power, they often compound methodological issues by aggregating studies that share similar limitations in how they measure and analyze criminological phenomena.

Most meta-analyses in criminology rely on transforming diverse findings into standardized effect sizes like Pearson's r or Cohen's d , typically in reference to a "nil" hypothesis of zero effect. While a technically sound method for statistically aggregating findings and avoiding subjective narrative reviews, as Meehl observed (1990b; 1990c), meta-analysis itself cannot overcome the fundamental interpretability problems that arise when original studies are based on vague theories and weak statistical tests. The statistical homogenization necessary for meta-analysis comes at the cost of the very precision and specificity needed for theoretical advancement.

Modern meta-analytic approaches do attempt to examine effect heterogeneity through meta-regression and subgroup analyses. However, these techniques typically measure variations in the magnitude of standardized effects rather than identifying fundamentally different patterns or relationships across contexts. This limitation stems not from meta-analysis itself but from the underlying studies that predominantly report simplified, standardized relationships rather than precise functional forms or context-specific patterns.

The methodological challenge is that meta-analysis can only be as good as the studies it synthesizes. When the primary literature consists largely of studies reporting directional associations and nil-hypothesis significance tests, even the most sophisticated meta-analytic techniques cannot transmute this into the kind of precise, context-specific knowledge needed for theoretical advancement. Meta-analysis, when applied to the typical criminological literature, may provide a more reliable estimate of the average effect size, but cannot overcome the fundamental limitations in how we conceptualize and measure relationships in the first place. Recognizing this, some meta-analysts in

criminology have moved beyond simple effect size aggregation. For instance, Pratt et al.'s (2010) meta-analysis of self-control theory went beyond reporting overall effects to examine theoretical contingencies and contextual factors that modify the strength and form of relationships, showing how meta-analysis can potentially contribute to theoretical refinement rather than merely box score tallying.

4.3. Misalignment Between Data Structure and Statistical Models

Beyond these historical constraints on our forms of knowledge, specific analytical practices within criminology frequently exacerbate the precision crisis. Consider the common practice of analyzing ordinal data using standard linear regression, or treating count data as if it were continuous and normally distributed. When researchers claim "substantively identical" results between models based solely on matching coefficient signs and significance thresholds, they sacrifice precision needed for robust theory development.

What is often meant by "substantively identical" is not that the models yield similar predictions or fit, but merely that the simplistic decision to declare an effect "significant" aligns, and the relationship points in the same direction. This analytical shortcut privileges the binary "box score" approach to knowledge over more precise understanding of the phenomena.

As Meehl (1967) argued in his critique of NHST in social science, relying solely on weakly directional significance tests is insufficient for building precise theories. The "crud factor" means vague directional predictions almost always find statistical support in large samples, making genuine falsification difficult. Some criminological work has recognized this limitation and employed analytical approaches better aligned with their data structures. For example, MacDonald and Lattimore's (2009) application of count models to criminal justice data demonstrates how appropriate modeling choices can reveal patterns in the data that would be obscured by standard linear models, while Wheeler, Worden and McLean's (2016) careful modeling of ordinal police behavior data shows how matching analytical approaches to data structures can yield more precise insights about the phenomena under study.

Yet, to put it bluntly, many popular criminological theories are so imprecise that our standard statistical tests of them are uninformative at worst and opaque descriptive statistics at best. This underscores the importance of what might be called "descriptive precision" in quantitative analysis,

especially in fields with developing theories. A half century ago, John Tukey (1969; quoted in Cohen, 1994: 86-87) encouraged psychologists to prioritize this:

“The physical sciences have learned much by storing up amounts, not just directions... Measuring the right things on a communicable scale lets us stockpile information about amounts. Such information can be useful, whether or not the chosen scale is an interval scale... these decades of progress would have been impossible had physicists and chemists refused either to record temperatures or to calculate with them.”

Applying Tukey’s principle to criminology means prioritizing analytical approaches that yield interpretable, precise descriptions of patterns appropriate to each data type, rather than focusing on standardized effect sizes that conceal more than they reveal. While various methodological approaches exist for enhancing analytical precision, adopting them alone is insufficient. Far more important than specific techniques is a fundamental rethinking of what constitutes valuable scientific knowledge in criminology.

This brings us to deeper challenges – Things We Don’t Understand – that go beyond methodological corrections to question the very forms of knowledge we produce and value. Even with improved analytical practices applied to conventional data, criminology would still face fundamental obstacles to achieving the kind of progressive problem shifts that characterize more cumulative sciences.

5. Things We Don’t Understand: Deeper Hurdles and the Pursuit of Progressive Problem Shifts

Even with better analytical practices, criminology faces more fundamental obstacles in achieving cumulative, predictive knowledge. These are the Things We Don’t Understand in a deeper sense, relating to theory construction, measurement, and the process of scientific discovery itself.

The urgency is underscored by persistent critiques of criminology’s status as a cumulative science. As Proctor and Niemeyer (2021) note, concerns from the 1933 Michael-Adler report about lacking reliable knowledge of crime’s causes remain relevant nearly a century later. Despite documenting countless correlates of crime, criminology has struggled to develop theoretical knowledge explaining why these correlations exist or resolving paradoxes from competing explanations. This contrasts with fields like physics or biology, which have achieved progressive problem shifts through

increasingly explanatory theories and precise predictions – Meehl’s (1990a) “money in the bank.”

5.1. From Description to Theory: The Missing Path in Criminology

Scientific progress in other fields reveals a pattern criminology has struggled to follow. As seen with Kepler’s laws and Mendel’s inheritance patterns (Section 2.8), precise empirical regularities preceded and constrained theoretical explanation. Theories were judged by their ability to account for these well-documented patterns with precision, creating an objective empirical foundation for evaluation.

Meehl (e.g., 1990b) articulated how social science often deviates from this pattern. Theories tend to be verbal rather than mathematical, leading to vague predictions. The “quasi-experimental modus operandi” combined with the “crud factor” makes it easy for vague predictions to appear significant in conventional statistical tests, meaning theories rarely face genuinely risky tests. Without falsification pressure, multiple competing theories persist indefinitely, none yielding precise novel predictions or non-ad-hoc explanations for anomalies.

In criminology, even our most robustly documented patterns like the age-crime curve or gender gaps are typically described qualitatively or as general trends. Typical theories provide plausible explanations but lack precision to predict exact quantitative shapes based on fundamental principles. Multiple contradictory theories claim consistency with these patterns because theoretical predictions and empirical descriptions lack sufficient precision to differentiate them rigorously.

Despite this general pattern, some criminological work has made progress in precisely documenting empirical regularities that constrain theory. For example, Weisburd’s (2015) “law of crime concentration” represents a step toward a Kepler-like empirical regularity in criminology – an attempt to make a risky prediction about a precise, quantitative pattern that, if robust, any viable theory of crime and place must be able to explain. Similarly, work on deterrence (Nagin & Pogarsky, 2001; Pogarsky, Roche, & Pickett, 2018) has documented not just that certainty matters more than severity (a directional claim) but specified how the relationship between perceived certainty and offending follows a particular functional form. Here, specification of mathematical models yielding testable predictions about non-linear relationships between punishment certainty and criminal behavior dialectically generate and emerge from rigorous empirical tests, with results driving theoretical refinements rather than post-hoc

explanations preserving original assumptions – a hallmark of progressive problem shifts. Beyond these, the broader field of crime science and environmental criminology has consistently sought to generate predictions about where and when crime will occur, leading to practical applications like geographic profiling and predictive policing, which rely on identifying precise spatial and temporal crime patterns to anticipate future events (cf. Rossmo & Wheeler, 2024; Brantingham & Brantingham, 1995).

A critical Thing We Don't Understand is how to identify and precisely describe fundamental building blocks of criminological reality – the robust phenomena (Eronen & Bringmann, 2021) – in a way that provides solid empirical foundation for theory building, particularly when our current form of knowledge (standardized effect sizes) is inadequate for capturing these phenomena in their full complexity.

5.2. Beyond Effect Sizes: The Limiting Force of Our Dominant Knowledge Form

At the heart of criminology's precision crisis lies our dominant form of scientific knowledge. Unlike the precise mathematical relationships or specific numerical ratios that characterize knowledge in progressive sciences (as detailed in Section 2.8), criminological knowledge typically takes the form of standardized effect sizes (correlations, regression coefficients, odds ratios) and their statistical significance.

This form of knowledge fundamentally limits our ability to develop and test precise theories for several reasons:

First, standardized effect sizes compress complex, potentially non-linear relationships into simplified metrics, discarding crucial information about functional forms, thresholds, and context dependencies. A correlation coefficient of 0.3 between self-control and offending tells us little about how this relationship might vary across different ranges of self-control, whether there are threshold effects, or how it might interact with environmental factors.

Second, this knowledge form is particularly ill-suited for adjudicating between competing theories. When multiple theories predict merely 'a positive relationship' between variables (as most do), finding such a relationship does little to differentiate between explanations. Without more precise predictions about magnitudes, functional forms, or context dependencies, empirical evidence struggles to drive theoretical refinement or replacement. For instance, the Eddington experiments were designed to

adjudicate between Newton's and Einstein's theories of gravity because they made quantitatively different and highly precise predictions about the bending of light around large objects like the sun, a crucial distinction beyond a mere directional effect (Strevens, 2020).

Third, standardized effect sizes are typically estimated in reference to a "nil" hypothesis of zero effect – a hypothesis that Meehl and others have argued is almost never precisely true in social science due to the crud factor. This focus on rejecting the nil hypothesis rather than testing specific theoretical predictions further weakens the connection between empirical findings and theoretical advancement.

Finally, the meta-analytic aggregation of these effect sizes, while addressing certain methodological concerns, compounds these limitations by further homogenizing findings across contexts and obscuring the very patterns that might drive theoretical development. When heterogeneity is addressed in meta-analysis, it typically examines variations in the magnitude of standardized effects rather than qualitatively different patterns or relationships.

The forms of knowledge we currently prioritize are artifacts of particular historical developments in statistics, as discussed in Section 4.1. Methods that simplified complex realities into standardized metrics were revolutionary when computational resources were limited, but have hardened into methodological orthodoxy that constrains how we conceptualize scientific evidence.

How we might develop alternative forms of knowledge more conducive to progressive problem shifts remains an open question. It's a fundamental Thing We Don't Understand. The challenge involves not just better statistical techniques but reimagining what constitutes valuable scientific evidence in criminology.

5.3. Towards Alternative Forms of Criminological Knowledge: Unresolved Challenges and Paths Forward

Having identified the limitations of standardized effect sizes as our dominant form of empirical knowledge, a crucial challenge is developing alternative forms of criminological knowledge more conducive to progressive problem shifts. Drawing inspiration from successful sciences, we propose several approaches that move beyond current practices to enable the kind of precise, falsifiable knowledge generation that drives cumulative science.

5.3.1. Prioritizing Precise Quantitative Description of Fundamental Empirical Regularities

This involves a fundamental shift in what we consider scientific knowledge. Rather than documenting correlations, this approach would focus on precise, quantitative descriptions of concrete criminological phenomena at the unit level—actual counts, rates, distributions, and patterns. Studying neighborhood crime, for example, would move beyond reporting partial correlations (e.g., $b = 0.4$) or average treatment effects and group-level heterogeneity to precise unit-level patterns, escaping the limited correlation paradigm.

A truly progressive approach would instead document precise, quantitative relationships at the unit level: How many burglaries, and of what types or characteristics, occur in neighborhoods with specific structural characteristics? What is the exact mathematical function that best describes the relationship between population density, guardianship measures, and robbery incidents? These questions seek the criminological equivalent of Kepler's laws – not merely correlations between variables, but risky predictions and, when quantifying, precise mathematical descriptions of how crime actually manifests across space, time, and social contexts (e.g., Rossmo & Wheeler, 2024).

Statistics, after all, is fundamentally about counting. In our pursuit of sophisticated analyses, we have often lost sight of the units being counted. Progressive sciences like physics and biology retain this connection – when Mendel documented a 3:1 ratio in inheritance patterns, he was counting actual peas with specific traits, not reporting a correlation coefficient between parental and offspring characteristics.

If criminological theories approached the precision of those in physics or biology, they would enable predictions at the unit level – the actual number of crimes expected in specific contexts given certain conditions. Deviations from these predictions would demand explanation, driving theoretical refinement. Imagine a theory that could predict not just that disadvantaged neighborhoods have “more crime” but that neighborhoods with specific characteristics would experience a precise number of crimes within a defined confidence interval. Deviations from these predictions would constitute meaningful anomalies requiring theoretical attention, rather than being dismissed as expected noise in a correlation.

Darwin's theory of evolution offers another instructive example. While not as mathematically

formalized as physics, Darwin's theoretical insights were built upon painstaking documentation of specific variations in actual organisms – not correlations between abstract variables. Scientific advances often follow from hyper-focus on detailing observable phenomena in their concrete manifestations (Strevens, 2020), not from premature theorizing based on simplified metrics that distance us from the very phenomena we seek to understand. Some promising work in this direction includes O'Brien's (2023) research on precisely quantifying and visualizing the temporal and spatial patterns of urban gun violence, documenting not merely that gun violence is concentrated, but exactly how it clusters in space and time, providing concrete phenomena that theories must explain.

5.3.2. A Critical Approach to Measurement Science

Our current knowledge form assumes that abstract constructs can be adequately represented through composite scores or latent variables derived from correlated indicators. This approach prioritizes internal consistency and unidimensionality over accurate representation of potentially complex constructs.

Moving beyond this requires understanding relationships between concrete indicators themselves and questioning whether standard psychometric models adequately represent complex traits in criminology (Revelle, 2024). Recent critiques highlight limitations of traditional measurement approaches. For example, Fried's (2017; Fried et al., 2020) work on depression challenges the assumption that symptom inventories measure a single underlying latent construct. Through network analysis and item-level examination, Fried demonstrates that specific symptoms have distinct etiologies, correlates, and consequences – information entirely lost when items are summed into a composite score.

This critique fundamentally challenges our dominant knowledge forms and encourages a standard higher than conventional psychometric indicators. Traditional metrics capture whether items “move together” but tell us little about whether items function identically in relation to external variables. True empirical justification would require meticulously describing the derivation chain anchoring observed phenomena to the theoretical construct for which each observation presumably is an indicator, then demonstrating that individual items show identical magnitudes of association with hypothesized causes and consequences, exhibit comparable functional forms, respond similarly to

experimental manipulations, and function precisely equivalently across different contexts.

Adopting this critical approach to measurement would transform how we conceptualize and operationalize key criminological constructs like self-control, social bonds, or strain. Some criminologists have begun to move in this direction. As examples of progressive work in this vein, Young and Weerman (2013) show how different measurement approaches lead to substantially different conclusions about peer influence processes, while Sweeten's (2012) critical examination of delinquency scales effectively demonstrates how different aggregation approaches can lead to substantially different substantive conclusions.

5.3.3. Rethinking Data Generation for Precision and Discovery

Our knowledge is constrained by both analytical practices and data types. Advancing beyond standardized effect sizes requires strategically generating data designed to reveal complex patterns and enable precise description. Criminal behavior demands data with temporal, spatial, and behavioral granularity beyond most datasets. For example, ecological momentary assessment reveals patterns invisible in traditional data, and street segment analysis demonstrates how increased spatial precision can reveal robust regularities (Weisburd, 2015; Chalfin, Kaplan, & Cuellar, 2021).

Fundamentally, criminology needs data collection explicitly designed to test competing theoretical predictions with precision. Rather than collecting data that can only support vague directional hypotheses, studies should be designed to capture specific patterns that would clearly distinguish between competing explanations. Patterson and Dishion's "deviancy training" work exemplifies this through direct observation and experimental manipulation (Dishion et al., 1996, 1999; Patterson et al., 2000). Skardhamar and Savolainen's (2014) test of the turning point hypothesis using monthly employment transition data and Deitzer and colleagues' (2025) analysis of hourly space-time budget data to assess situational peer effects within hours after exposure to peers offer further exemplars.

Assuming we move beyond cookbook statistical modeling and correlational or mean-difference summaries of resulting data, these approaches hold potential for generating forms of knowledge that go beyond standardized effect sizes to capture the dynamic, contextual, and complex nature of

criminological phenomena – providing a stronger foundation for theoretical advancement.

5.3.4. Developing and Testing Computational Models of Theories

Perhaps the most promising avenue for generating new forms of criminological knowledge involves developing formal computational models that instantiate theoretical claims in precise algorithms and parameters. Unlike our current approach of testing vague theoretical implications against simplified metrics, computational modeling forces theories to be precisely specified and enables rigorous comparison between theoretical predictions and empirical observations.

Consider how computational modeling might transform a theory like General Strain Theory (Agnew, 1992). Rather than simply testing whether strain correlates with delinquency, a computational model would specify precisely how different types of strain (e.g., parental rejection; financial hardship) are processed, how they interact with coping resources (e.g., social support; self-efficacy) and social controls, and how these processes unfold over time to produce specific behavioral outcomes. For instance, in generating such a model, one might predict that 'a maximal increase from "none" to "very high" on an ordinal item measuring *perceived financial strain*, when an individual's reported *self-efficacy for conventional problem-solving* is below a threshold of "3" (on a 5-point Likert-type item), will change the expected proportion reporting *intent to engage in property offenses* in the next year from approximately zero individuals (0.0 to 0.01) to about half the population (0.45 to 0.55) among adolescents aged 14-16, with this effect peaking within six months of the strain's onset.'

Such precision typically cannot be derived from existing criminological theory, yet it is essential for simulating data from computational models. Of course, such probability-based predictions about relationships between concepts themselves should not be viewed as the ultimate form of scientific knowledge as they depend upon complex logical derivation chains (Meehl, 1990a) and mechanistic assumptions (Proctor & Niemeyer, 2021), and their very form as probability predictions reflects the uncertainty inherent in using a highly simplified model to predict outcomes that occur within a complex web of interactions among other joint causal forces that aggregately suppress or increase the outcome's occurrence. Still, moving toward this level of specificity transforms vague associations into falsifiable predictions at the unit level. This model could then generate synthetic data that could be

compared directly with empirical observations. Discrepancies would highlight specific aspects of the theory requiring refinement, driving progressive problem shifts focused on particular mechanisms rather than the theory as a whole.

This approach forces theoretical precision and provides a bridge between abstract conceptualization and empirical reality. It transforms knowledge from simplified effect sizes to dynamic, process-oriented representations that capture the complexity of criminological phenomena while enabling risky predictions and rigorous theory testing. Similar approaches have dramatically advanced fields like cognitive psychology (Sun, 2008) and could produce comparable benefits for criminological theory.

These computational models need not be mathematically complex or require advanced programming skills to develop. Even simple agent-based models can force theoretical precision by requiring explicit specification of mechanisms and parameters, thereby generating more risky predictions than conventional verbal theories (Johnson & Groff, 2014). Recent work by Birks, Townsley, and Stewart (2012) demonstrates how agent-based modeling can formalize criminological theories about offender decision-making and environmental influences, generating precise predictions that can be tested against empirical data. Similarly, Brantingham and Tita (2008) shows how computational modeling can derive crime patterns from first principles about offender decision-making, creating a bridge between abstract theory and concrete empirical patterns.

Pursuing these shifts requires embracing methodological pluralism and potentially “anything goes” creativity in discovery (Feyerabend, 1975), but explorations must eventually lead back to rigorous empirical documentation and testable claims about observable reality.

6. Critiques and Barriers to Progressive Problem Shifts

Before outlining the path forward, we address critical perspectives and practical impediments to achieving greater precision in criminological theory and method. These range from fundamental disagreements about the nature of social phenomena to the pragmatic realities of academic incentive structures that actively discourage many of the practices we propose. Indeed, it may be that deeper disciplinary tendencies might persist even if incentives were reformed.

6.1. The Nature of Criminological Phenomena: Complexity and Appropriateness of Standards

A primary objection to our call for greater precision is that human behavior, particularly criminal behavior, is fundamentally too complex, contingent, and context-dependent to yield the kind of precise, quantitative theories seen in physics or certain areas of biology. Some might see applying “hard science” standards to criminology as representing a form of physics envy that fails to respect the unique nature of social phenomena. We acknowledge that criminology studies phenomena embedded in historical, cultural, and institutional contexts that may preclude the discovery of universal, time-invariant laws akin to Newton’s gravitation. However, acknowledging this complexity does not justify settling for imprecision where greater precision is possible. Our argument is that this allowable precision is considerably higher than current practice typically achieves. Complexity itself can be precisely described, as seen in meteorology, which deals with chaotic systems yet has dramatically improved its predictive capacity through precise measurement, computational modeling, and explicit uncertainty quantification. Similarly, criminology can aspire to more precise description of complex patterns and more explicit modeling of uncertainty without claiming deterministic prediction of individual behavior.

Critics might also question whether our examples from physics and biology are appropriate as analogies for criminology, given the agency of human subjects. However, this objection misses our central point: we draw from these examples not their specific content, but the forms of knowledge they represent – knowledge that is appropriate to their subject matter. We advocate for criminology to develop forms of knowledge appropriate to the complexity of crime, but without defaulting to the least precise metrics merely because they are conventional.

6.2. Methodological Pluralism and Practical Constraints

Others might object that focusing on precision and quantification risks overlooking the important qualitative, interpretive dimensions of crime and criminal justice. We agree that quantitative precision should complement, not replace, rich qualitative understanding. Indeed, precise quantitative description often generates puzzles that qualitative research is uniquely positioned to explain, while qualitative insights can suggest new constructs or relationships requiring quantitative validation. The progressive problem shifts we advocate should incorporate methodological pluralism while

maintaining rigorous standards appropriate to each approach.

Despite its theoretical appeal, computational modeling might be viewed by some as impractical given the complexity of social phenomena. Models necessarily simplify reality, and poorly specified models might lead to misleading conclusions. This is a valid concern, but it argues for careful model development rather than abandoning the approach. Even simple models can force theoretical precision by requiring explicit specification of mechanisms and parameters. Models should be judged not by whether they capture all aspects of reality, but by whether they advance our understanding beyond what conventional approaches provide (see Smaldino, 2023).

A final objection might concern resource constraints – that granular data collection and complex modeling are too expensive or time-consuming. While acknowledging practical limitations, we suggest that more precise theory development itself can help prioritize limited resources by generating specific, testable predictions that narrow the empirical focus. In essence, precision in theory and method is not just scientifically desirable but may also prove more efficient in the allocation of limited research resources over time.

6.3. Academic Incentive Structures and Disciplinary Tendencies

While the conceptual and methodological objections addressed above warrant consideration, another significant practical barrier, or at least a substantial compounding factor, lies in the academic incentive structures that are at odds with many of the practices we propose. Current systems disproportionately reward the publication of novel, statistically significant findings in high-impact journals, potentially discouraging the slower, painstaking work of precise description, rigorous replication, and the transparent reporting of null results from risky tests. These incentives may directly reinforce the standardized effect size paradigm: researchers are rewarded for producing clean, simplified metrics of association that can be easily summarized and published, rather than complex, detailed descriptions of criminological phenomena that might better advance theoretical understanding.

This publication bias creates a perverse incentive where researchers are rewarded for producing statistically significant findings, regardless of the precision of the evidence or the riskiness of the test. Under such conditions, it is rational for researchers to prefer vague theories that can be ‘supported’ by almost any data pattern, to employ flexible analytical

practices that can extract significant results from noise, and to selectively report analyses that yield desired outcomes. These practices, while individually understandable responses to existing incentives, collectively undermine progressive problem shifts. Moreover, these incentives may perpetuate and *even be symptoms of* a deeper culture of tolerance for imprecision, potentially masking more fundamental intellectual habits or preferences within the discipline itself, suggesting that the problem isn't *just* external systemic pressures. Indeed, according to some arguments, social sciences like criminology may face a persistent challenge in attracting individuals with the drive and skillset to generate a rigorous theoretical science, instead drawing those more motivated by social change or practical application, which can inadvertently favor broader narratives over accumulation precise, falsifiable knowledge (cf. Blalock, 1994; Turner, 1994; Savolainen, 2024).

Several specific features of current incentive structures that might impede the development of alternative forms of knowledge or that perhaps reflect deeper cultural barriers:

- *A premium on novelty* discourages replication and the detailed work needed to document precise empirical regularities, leaving the empirical foundation perpetually unstable. This emphasis may stem from a disciplinary culture valuing continuous “newness” over foundational consolidation.
- *An emphasis on theoretical framing* often forces descriptive findings into existing frameworks, hindering challenges to those frameworks and leading to premature theorizing. This could reflect a preference for fitting data to established narratives rather than letting precise empirical patterns drive theoretical development.
- *A bias against null findings* discourages rigorous, risky tests essential for falsification and progress, as careers depend on ‘successful’ studies. This might be a symptom of a disciplinary discomfort with genuine refutation or a lack of emphasis on falsifiability (i.e., of risky predictions and not “nil” hypotheses) in training.
- *A reward for quantity over precision* encourages numerous studies with minimal investment in precise measurement or detailed description, disincentivizing time-intensive approaches like computational modeling and high-resolution data collection. This could reflect a cultural prioritization of

productivity metrics over deep, rigorous inquiry.

- *Publication formats constrain complexity*, privileging brief reports of simplified metrics over rich, detailed, open, transparent, and reproducible accounts of phenomena. This practical constraint might be passively accepted due to a disciplinary comfort with less demanding forms of evidence.

Cumulatively, these features distort the scientific record and may lead to faux empirical support for a theory. Many theories in criminology thus generate “monopoly money in the bank” or seeming empirical support without rigorous testing of core theoretical implications. This imprecision allows advocates to point to a multitude of studies claiming support, even when such support is weak or untestable.

Multiple scholars have proposed individualized and institutional reforms, including pre-registration, registered reports, altered promotion criteria, and new publication formats. Yet, it is noteworthy that fields like physics, despite also operating within incentive structures valuing publication and novelty, have nonetheless achieved significant progressive problem shifts. This might suggest that deeper disciplinary characteristics, such as a fundamental commitment to risky tests, precise prediction, and falsifiability, may be more important than incentive structures (cf. Yarkoni, 2023). If individuals with such a foundational scientific mindset were to approach criminological phenomena *en masse*, it is plausible that greater progress would ensue regardless of the current incentive landscape. While changing entrenched incentive structures and widespread practices presents significant challenges and may not resolve all disciplinary issues on its own, the history of science suggests that fields and even individuals capable of reforming their practices to enable progressive problem shifts ultimately can produce more valuable and lasting knowledge. The criminological examples cited throughout this article, and many others, illustrate criminology’s potential for scientific progress in precision, granular data collection, rich description, computational modeling, and severe testing, even if current statistical knowledge forms still limit its cumulative potential.

8. Conclusion: Beyond the Box Score - Toward a Progressive Criminological Science

The precision crisis in criminological science represents both a challenge and an opportunity. By

recognizing how our very conception of scientific knowledge has been constrained by historically contingent statistical practices, we can identify concrete areas where improvement is possible at methodological, theoretical, and institutional levels.

The path forward requires commitment to several key principles: moving beyond standardized effect sizes and significance testing as the primary currency of scientific knowledge; prioritizing precise description of concrete phenomena before abstract theorizing; subjecting theories to genuinely risky tests rather than weak confirmations; approaching measurement with critical scrutiny rather than uncritical application of conventional psychometrics; generating high-resolution data specifically designed to differentiate between competing explanations; developing and testing formal computational models of theories; and reforming academic incentives to reward these practices. Furthermore, fostering a genuine commitment to scientific progress might require more robust undergraduate and graduate training in the history and philosophy of science, explicitly exploring historical examples of precision and progressive problem shifts.

While we do not suggest that criminology can or should become physics, we maintain that the field can achieve considerably greater precision and cumulativeness than current practices yield. The examples of success we have highlighted demonstrate that progress is possible even within the constraints of studying complex human behavior.

As criminology continues to evolve, its scientific maturity will ultimately be judged not by tallies of significant findings or meta-analytic summaries of standardized effect sizes, but by its ability to generate increasingly precise answers to fundamental questions about crime and its causes – answers that withstand rigorous attempts at falsification and provide genuine guidance for both theory and practice. By confronting the precision crisis directly and reimagining the very forms of knowledge we produce, criminology can move closer to fulfilling its potential as a cumulative scientific enterprise capable of building “money in the bank” through progressive problem shifts that expand our understanding of crime and its causes. Ultimately, a more precise criminological science is not merely an academic ideal; it is a prerequisite for developing genuinely evidence-based policies and interventions that can effectively address crime and its complex causes in the real world.

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